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| **Fiscal Year:** | FY 2005 | **Task Last Updated:** | FY 07/19/2018 |
| **PI Name:** | Reschke, Millard F Ph.D. | | |
| **Project Title:** | Modification of Eccentric Gaze-Holding | | |
| **Division Name:** | Human Research | | |
| **Program/Discipline:** | NSBRI Teams | | |
| **Program/Discipline--Element/Subdiscipline:** | NSBRI Teams--Sensorimotor Adaptation Team | | |
| **Joint Agency Name:** | | **TechPort:** | No |
| **Human Research Program Elements:** | None | | |
| **Human Research Program Risks:** | None | | |
| **Space Biology Element:** | None | | |
| **Space Biology Cross-Element Discipline:** | None | | |
| **Space Biology Special Category:** | None | | |
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| **City:** | Houston | **State:** | TX |
| **Zip Code:** | 77058-3607 | **Congressional District:** | 36 |
| **Comments:** | | | |
| **Project Type:** | Ground | **Solicitation / Funding Source:** | NSBRI |
| **Start Date:** | 08/01/2001 | **End Date:** | 09/30/2005 |
| **No. of Post Docs:** | 1 | **No. of PhD Degrees:** | |
| **No. of PhD Candidates:** | 0 | **No. of Master' Degrees:** | |
| **No. of Master's Candidates:** | 0 | **No. of Bachelor's Degrees:** | |
| **No. of Bachelor's Candidates:** | 0 | **Monitoring Center:** | NSBRI |
| **Contact Monitor:** | | **Contact Phone:** | |
| **Contact Email:** | | | |
| **Flight Program:** | | | |
| **Flight Assignment:** | | | |
| **Key Personnel Changes/Previous PI:** | | | |
| **COI Name (Institution):** | | | |
| **Grant/Contract No.:** | NCC 9-58-NA.002.08 | | |
| **Performance Goal No.:** | | | |
| **Performance Goal Text:** | | | |
| | Clear vision and accurate localization of objects in the environment are prerequisites for reliable performance of motor tasks. Space flight confronts the crew member with a stimulus rearrangement that requires adaptation to function effectively with the new requirements of altered spatial orientation and motor coordination. Adaptation and motor learning driven by the effects of cerebellar disorders may share some of the same demands that face our astronauts. One measure of spatial localization shared by the astronauts and those suffering from cerebellar disorders that is easily quantified, and for which a neurobiological substrate has been identified, is the control of the angle of gaze (the "line of sight"). The disturbances of gaze control that have been documented to occur in astronauts, both in-flight and postflight, can be directly related to changes in the extrinsic gravitational environment and intrinsic proprioceptive mechanisms thus, lending themselves to description by mathematical models. The basic models can be formulated using normal, non-astronaut test subjects and subsequently extended using centrifugation techniques to alter the gravitational and | | |

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| **Task Description:** | propriocep-tive environment of these subjects. Further tests and extensions of the models can be made by studying abnormalities of gaze control in patients with cerebellar disease. Furthermore, we are proposing a simple approach to the problem of adaptation monitoring and determination of countermeasurement effectiveness. The specific aims of this study are: (1) To investigate the mechanisms of gaze-holding in normal, non-astronaut subjects, with the head held in various orientations with respect to gravity and the head held in various orientations relative to both gravity and the trunk. This will involve characterizing the time constant of centripetal gaze drift, the rate in which the eyes naturally drift back toward the null position following an eccentric eye movement. (2) To investigate the mechanisms that adaptively compensate for gaze-holding failure, especially the "rebound nystagmus" phenomenon, which decreases the rate of centripetal drift of the eyes. We will study the time course of rebound nystagmus in normal, non-astronaut subjects. (3) To investigate the stimulus rearrangement and adaptation resulting from exposure to gravitoinertial environments greater that 1G using prolonged exposure to centrifugation. We will compare measures of the time constant of centripetal drift pre-, in-, and postcentrifugation. (4) To study mechanisms that adaptively compensate for gaze-holding failure in patients with vestibular cerebellar disease who show impaired gaze-holding ability. We will compare gaze-holding defects and rebound nystagmus in patients with those induced in our normal subjects. |
| **Rationale for HRP Directed Research:** | |
| **Research Impact/Earth Benefits:** | NASA, jointly with the NSBRI has established a critical path along which all future research must follow to ensure the ultimate goal of developing scientifically-based countermeasure against the neurovestibular problems associated with space flight. Specifically, five major neurovestibular risk areas have been identified. This proposal address three of the five areas on the critical path roadmap: a) measurement of disorientation and consequential reduced performance on cognitive and physical tasks, including vehicle egress, acute and/or spontaneous and head movement contingent vertigo, nystagmus, oscillopsia, saccadic errors, and reduced visual acuity; b) impaired cognitive and/or physical performance due to spatial disorientation, motion sickness symptoms, or treatments as a result of changes in g-level, or use of artificial gravity; and c) measurement of potential permanent impairment of orientation and balance function due to microgravity or radiation that may cause chronic imbalance, gait ataxia, vertigo, chronic vestibular insufficiency, and poor visual performance. Currently there is no simplified, easily administered inflight or postflight test that can be used to evaluate either the above identified risks or the efficacy of newly developed countermeasures. Importantly, neither the preflight or postflight neurological exams that have been administered throughout the history of the U.S. space program have been tailored or sharpened to identify specific changes associated with space flight. NASA has identified, as a part of its critical path, the requirement for specific, simplified tests that can assist in the identification and tracking of neurovestibular changes associated with the transition between 1-g and 0-g, 0-g and partial-g, or 0-g and artificial gravity with a minimum of impact on crew and mission time or resources. This research, based on previous observations, will develop a simplified test that meets the requirements of the critical path and NASA’s operational goals. Importantly, this research, by its very nature of helping NASA develop a critical test of sensory-motor function will also assist the medical community develop a simple test of vestibular function that can be easily administered. Second to this benefit is the possibility that the gaze-holding test developed with this research will be instrumental in assisting in the construction of non-drug related countermeasures for motion sickness and locomotor problems. |
| **Task Progress:** | Ed. note 7/19/2018--Reporting not received. Submitted at this time to include publication resulting from the work. |
| **Bibliography Type:** | Description: (Last Updated: 11/14/2025) |
| **Articles in Peer-reviewed Journals** | Schneider R, Liao K, Walker MF, Bronstein A, Leigh RJ. "Behavior of the human translational vestibulo-ocular reflex during simultaneous head translation and rotation." J Vestib Res. 2014;24(5-6):329-33. https://doi.org/10.3233/VES-140522 ; PubMed PMID: 25564074 , May-2014 |
| **Articles in Peer-reviewed Journals** | Clément G, Beaton KH, Reschke MF, Wood SJ. "Effects of motion paradigm on human perception of tilt and translation." Sci Rep. 2022 Jan 26;12(1):1430. https://doi.org/10.1038/s41598-022-05483-6 ; PMID: 35082357; PMCID: PMC8792002 , Jan-2022 |